Practice workbook · Two constants from factor and remainder information

Forming two equations in two unknown coefficients from factor and remainder information (sometimes about f'(x) or after integrating) and solving them simultaneously.

13 question parts: 1 worked-example retry and 12 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-IDN-09

1. Question 5 · part a · 5 marks

2022 JAN · Paper 01

Worked-example retry · Attempt part a only.

5 $f(x) = ax^3 + 5bx^2 + 8ax - 4b$ where a and b are integers

Given that

$(x + 2)$ is a factor of $f(x)$

and that

when $f(x)$ is divided by $(x + 3)$ the remainder is 21

(a) show that $a = 2$ and find the value of b

(5)

(b) Use algebra to solve the equation $f(x) = 0$

(4)

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1. Question 5 · part a · 5 marks · Working and answer

Working and answer

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2. Question 4 · part a.i · 0 marks

2023 JAN · Paper 01

Further practice · Attempt part a.i only.

4 $f(x) = x^3 + px^2 + qx + 6$ where p and q are constants.

Given that $(x - 1)$ is a factor of $f(x)$ and that when $f(x)$ is divided by $(x + 1)$ the remainder is 8

(a) (i) show that $p = -2$

(ii) find the value of q

(6)

(b) Hence, solve the equation $f(x) = 0$

(3)

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2. Question 4 · part a.i · 0 marks · Working and answer

Working and answer

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3. Question 4 · part a.ii · 6 marks

2023 JAN · Paper 01

Further practice · Attempt part a.ii only.

4 $f(x) = x^3 + px^2 + qx + 6$ where p and q are constants.

Given that $(x - 1)$ is a factor of $f(x)$ and that when $f(x)$ is divided by $(x + 1)$ the remainder is 8

(a) (i) show that $p = -2$

(ii) find the value of q

(6)

(b) Hence, solve the equation $f(x) = 0$

(3)

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3. Question 4 · part a.ii · 6 marks · Working and answer

Working and answer

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4. Question 2 · part a · 6 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part a only.

2 $f(x) = x^3 + px + q$ where p and q are constants.

The remainder when $f(x)$ is divided by $(x - 1)$ is -12

The remainder when $f(x)$ is divided by $(x - 4)$ is 30

(a) Find the value of p and the value of q.

(6)

Using your values of p and q

(b) show that $f(3) = 0$

(1)

(c) Express $f(x)$ as a product of linear factors.

(3)

(d) Hence solve the equation $f(x) = 0$

(1)

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4. Question 2 · part a · 6 marks · Working and answer

Working and answer

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5. Question 5 · part a · 4 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

5 $f(x) = x^3 + x^2 + x + c$ where c is a constant

The remainder when $f(x)$ is divided by $(x-2)$ is 4 times the remainder when $f(x)$ is divided by $(x+1)$

(a) Show that $c = 6$

(4)

Given that $(x+2)$ is a factor of $f(x)$

(b) show that the equation $f(x) = 0$ has only one real root.

(4)

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5. Question 5 · part a · 4 marks · Working and answer

Working and answer

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6. Question 4 · part a · 6 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

4 $f(x) = 2x^3 + ax^2 + bx + 18$ where a and b are constants

When $f'(x)$ is divided by $(x - 2)$ the remainder is 5

Given that $(x - 2)$ is a factor of $f(x)$

(a) find the value of a and the value of b.

(6)

(b) Express $f(x)$ as a product of linear factors.

(3)

(c) Hence use algebra to solve the equation $f(x) = 0$

(2)

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6. Question 4 · part a · 6 marks · Working and answer

Working and answer

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7. Question 5 · part a · 5 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

5 $f(x) = 2x^3 + ax^2 - 14x + b$ where a and b are constants.

When $f(x)$ is divided by $(x - 4)$ the remainder is 39

Given that $(x - 1)$ is a factor of $f(x)$

(a) show that $a = -3$ and find the value of b

(5)

(b) Hence factorise $f(x)$ completely.

(4)

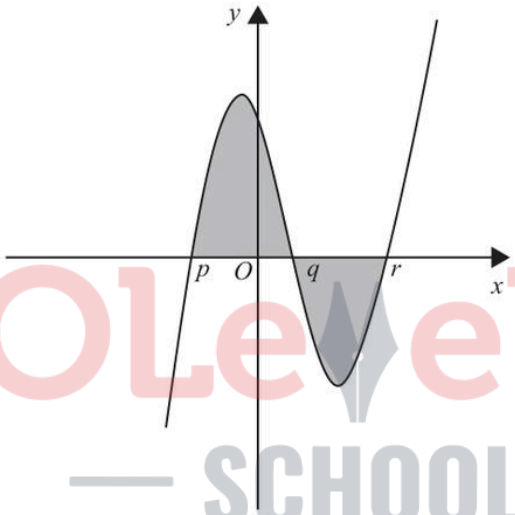


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows part of the curve C with equation $y = f(x)$

Given that C crosses the x-axis at the points with coordinates $(p, 0)$, $(q, 0)$ and $(r, 0)$

(c) write down the value of p, the value of q and the value of r

(3)

The region shown shaded in Figure 3 is bounded by the curve and the x-axis.

(d) Use algebraic integration to find the exact area of the shaded region.

(4)

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7. Question 5 · part a · 5 marks · Working and answer

Working and answer

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8. Question 4 · part a.i · 0 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part a.i only.

4 $f(x) = px^3 + qx^2 - 37x - 12q$ where p and q are constants.

When $f'(x)$ is divided by $(x + 2)$ the remainder is -33

Given that $(x + 5)$ is a factor of $f(x)$

(a) (i) show that $p = 2$

(ii) find the value of q

(6)

(b) Hence, use algebra to factorise $f(x)$ completely.

(3)

(c) Hence solve the equation $f(x) = 0$

(2)

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8. Question 4 · part a.i · 0 marks · Working and answer

Working and answer

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9. Question 4 · part a.ii · 6 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part a.ii only.

4 $f(x) = px^3 + qx^2 - 37x - 12q$ where p and q are constants.

When $f'(x)$ is divided by $(x + 2)$ the remainder is -33

Given that $(x + 5)$ is a factor of $f(x)$

(a) (i) show that $p = 2$

(ii) find the value of q

(6)

(b) Hence, use algebra to factorise $f(x)$ completely.

(3)

(c) Hence solve the equation $f(x) = 0$

(2)

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9. Question 4 · part a.ii · 6 marks · Working and answer

Working and answer

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10. Question 7 · part a · 5 marks

2025 NOV · Paper 02

Further practice · Attempt part a only.

7 $f(x) = 2x^3 + Ax^2 + Bx - 20$ where A and B are constants.

Given that

- $(2x+1)$ is a factor of $f(x)$
- when $f'(x)$ is divided by $(x-1)$ the remainder is -27

(a) show that $A = 3$ and find the value of B

(5)

(b) Hence, using algebra, solve the equation $f(x) = 0$

(3)

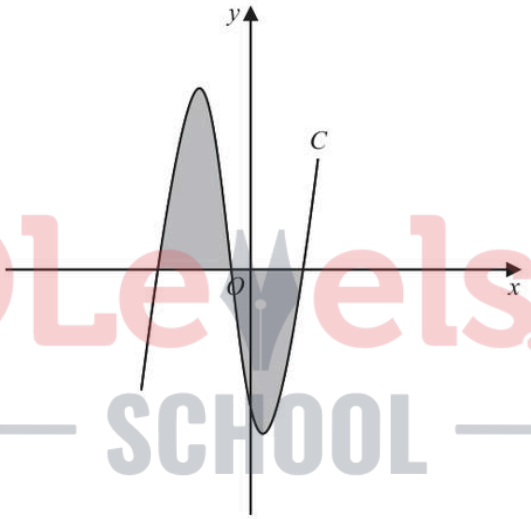


Figure 2

Figure 2 shows a sketch of part of the curve C with equation $y = f(x)$

(c) Use calculus to find the exact area of the finite region bounded by C and the x-axis, shown shaded in Figure 2

(4)

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10. Question 7 · part a · 5 marks · Working and answer

Working and answer

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11. Question 11 · part a · 6 marks

2021 NOV · Paper 01

Further practice · Attempt part a only.

11 $f'(x) = ax^2 - 14x - 10$ where $a \in \mathbb{Z}$

Given that $(x - 4)$ is a factor of $f(x)$ and that when $f(x)$ is divided by $(x + 1)$ the remainder is 25

(a) show that $a = 6$

(6)

(b) Hence use algebra to solve the equation $f(x) = 0$

(6)

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11. Question 11 · part a · 6 marks · Working and answer

Working and answer

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12. Question 4 · part a · 5 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

4 $f(x) = x^3 + px^2 + qx + 7$ where p and q are integers.

$(x + 1)$ is a factor of $f(x)$
The remainder when $f(x)$ is divided by $(x + 2)$ is -5

(a) Find the value of p and the value of q

(5)

(b) Hence, show that $f(x) = 0$ has only one real root.

(3)

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12. Question 4 · part a · 5 marks · Working and answer

Working and answer

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13. Question 5 · part a · 5 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

5 $f(x) = 2ax^3 + x^2 - bx + 3a$ where a and b are integers.

Given that $(x + 2)$ and $(x - 1)$ are both factors of $f(x)$

(a) show that $a = 2$ and find the value of b

(5)

(b) Hence factorise $f(x)$ completely.

(2)

Hence, given that $h(y) = 2^{(3y+2)} + 2^{2y} - 11(2^y) + 6$

(c) solve the equation $h(y) = 0$

Where appropriate give your answers to 3 decimal places.

(5)

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13. Question 5 · part a · 5 marks · Working and answer

Working and answer

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